Final Exam Review

1) Evaluate $\int x \cdot \sqrt{x^2 - 100} \cdot dx$ Hint: Let $u = x^2 - 100$

a) Write $\int x \cdot \sqrt{x^2 - 100} \cdot dx$ in terms of u ________________________

b) $\int x \cdot \sqrt{x^2 - 100} \cdot dx =$ ________________________________

Solution:

1 $\int x \cdot \sqrt{x^2 - 100} \cdot dx$

1 $u = x^2 - 100$

$\frac{du}{dx} = 2x$

$du = 2x \cdot dx$

$\frac{1}{2} du = \frac{1}{2} \cdot 2x \cdot dx = x \cdot dx$

a) $\int x \cdot \sqrt{x^2 - 10} \cdot dx = \int \sqrt{x^2 - 10} \cdot x \cdot dx$

$= \int \sqrt{u} \cdot \frac{1}{2} du = \frac{1}{2} \cdot \int u^{1/2} \cdot du$

b) $\frac{1}{2} \int u^{1/2} \cdot du = \frac{1}{2} \left(u^{3/2}\right) \cdot \frac{2}{3}$

$= \frac{1}{3} \left(x^2 - 100\right)^{3/2}$

$\int x \cdot \sqrt{x^2 \; 100} \cdot dx = \frac{1}{3} \left(x^2 - 100\right)^{3/2} + C$

2) Evaluate $\int \frac{x}{x^2-64} \cdot dx$ Hint: Let $u = x^2 - 64$

a) Write $\int \frac{x}{x^2-64} \cdot dx$ in terms of u ____________________________

b) $\int \frac{x}{x^2-64} \cdot dx =$ ____

Solution:

(2) $\int \frac{x}{x^2-64} \cdot dx$

$$u = x^2 - 64$$
$$du/dx = 2x$$
$$du = 2x \cdot dx$$
$$\tfrac{1}{2} du = \tfrac{1}{2} \cdot 2x \, dx$$
$$\tfrac{1}{2} du = x \cdot dx$$

(a) $\int \frac{x}{x^2-64} \cdot dx = \int \frac{1}{x^2-64} \cdot x \, dx$

$$= \int \frac{1}{u} \cdot \tfrac{1}{2} du$$
$$= \tfrac{1}{2} \int \frac{1}{u} \cdot du$$
$$= \tfrac{1}{2} \ln|u|$$

(b) $= \tfrac{1}{2} \ln|x^2-64| + C$

3) Evaluate $\int 5x \cdot \sqrt{x-7} \cdot dx$ Hint: Let $u = x - 7$

a) Write $\int 5x \cdot \sqrt{x-7} \cdot dx$ in terms of u ________________________

b) $\int 5x \cdot \sqrt{x-7} \cdot dx =$ ______________________________________

Solution:

(3) $\int 5x \cdot \sqrt{x-7} \cdot dx$

a) let $u = x - 7 \Rightarrow x = u + 7$

$du/dx = 1$

$du = dx$

$\int 5x \cdot \sqrt{x-7} \cdot dx = 5\int (u+7) \cdot \sqrt{u}\, du$

$= 5\int (u^{3/2} - 7u^{1/2}) \cdot du$ Note: $\sqrt{u} = u^{1/2}$

$= 5\left[\dfrac{u^{5/2}}{5/2} - \dfrac{7u^{3/2}}{3/2}\right]$

$= 2u^{5/2} - \frac{70}{3}u^{3/2}$

(b) $= 2(x-7)^{5/2} + \frac{70}{3}(x-7)^{3/2} + C$

4) Evaluate $\int x^3 \cdot \ln x \cdot dx$ Hint: Use Integration by Parts

Let $u = \ln x$ and $dv = x^3 \cdot dx$

a) $v =$ ______________________________

b) $\int x^3 \cdot \ln x \cdot dx =$ ____

Solution:

④ $\int x^3 \cdot \ln x \cdot dx$

$u = \ln x$ $\qquad$ $dv = x^3 \cdot dx$

$\frac{du}{dx} = \frac{1}{x}$ $\qquad$ $\int 1 \cdot dv = \int x^3 \cdot dx$

$du = \frac{1}{x} \cdot dx$

ⓐ $\qquad V = x^4/4$

$$\int u \cdot dv = u \cdot v - \int v \cdot du$$

$$\int x^3 \cdot \ln x \cdot dx = (\ln x) \cdot (x^4/4) - \int \frac{x^4}{4} \cdot \frac{1}{x} dx$$

$$= \frac{1}{4}(\ln x) x^4 - \int \frac{1}{4} \cdot x^3 \cdot dx$$

$$= \frac{1}{4}(\ln x) \cdot x^4 - \frac{1}{4}(x^4/4)$$

ⓑ $$= \frac{1}{4}(\ln x) \cdot x^4 - \frac{1}{16} x^4 + C$$

5) Evaluate $\int \frac{\sqrt{x^2-9}}{x} \cdot dx$ Hint: Use Table of Integrals

$\int \frac{\sqrt{x^2-9}}{x} \cdot dx =$ ______________________________

Solution:

Formula 53:

$$\int \frac{\sqrt{u^2-a^2}}{u} du = \sqrt{u^2-a^2} - a\sec^{-1}\left(\frac{u}{a}\right) + C$$

Let $u^2 = x^2$ and $a^2 = 9$

$u = x$ $a = 3$

$du / dx = 1$

$du = dx$

$$\int \frac{\sqrt{x^2-9}}{x} dx = \int \frac{\sqrt{u^2-a^2}}{u} du = \sqrt{u^2-a^2} - a\sec^{-1}\left(\frac{u}{a}\right) + C$$

$$= \sqrt{x^2-9} - 3\sec^{-1}\left(\frac{x}{3}\right) + C$$

6) Find $\int \frac{5x-2}{x^2-x} \cdot dx$

a) Decompose $\frac{5x-2}{x^2-x} =$ ______________________________

b) $\int \frac{5x-2}{x^2-x} \cdot dx \quad =$ ______________________________

Solution:

(6) $\int \frac{5x-2}{x^2-x} \cdot dx$

a) $\frac{5x-2}{x^2-x} = \frac{5x-2}{x\cdot(x-1)} = \frac{A}{x} + \frac{B}{x-1}$

$\Rightarrow \frac{5x-2}{x\cdot(x-1)} = \frac{A(x-1)+Bx}{x\cdot(x-1)}$

$\Rightarrow 5x-2 = A(x-1) + B\cdot x$

Let $x=1$: $5(1)-2 = A(0) + B(1)$

$3 = B$

Let $x=0$: $5(0)-2 = A(-1) + B(0)$

$-2 = -1A$

$A = 2$

(b) $\int \frac{5x-2}{x^2-x} dx = \int \left(\frac{2}{x} + \frac{3}{x-1}\right) dx$

$= 2\int \frac{1}{x} dx + 3\cdot\int \frac{1}{x-1} dx$

$= 2\cdot \ln|x| + 3\cdot \ln|x-1| + C$

7) Find $\int \frac{15x}{(3+7x)^2} \cdot dx$ Hint: Use Table of Integrals

$$\int \frac{15x}{(3+7x)^2} \cdot dx = \underline{\hspace{8cm}}$$

Solution:

Formula 29: $\int \frac{u}{(a+bu)^2} du = \frac{1}{b^2}\left[\frac{a}{a+bu} + \ln|a+bu|\right] + C$

Let $u = x$ $\quad a = 3$ $\quad b = 7$.

$du/dx = 1$

$du = dx$

$$\int \frac{15x}{(3+7x)^2} dx = 15\int \frac{x}{(3+7x)^2} dx = 15\int \frac{u}{(a+bu)^2} du$$

$$= 15\left[\frac{1}{7^2} \cdot \left(\frac{3}{3+7x} + \ln|3+7x|\right)\right] = \frac{15}{49} \cdot \left(\frac{3}{3+7x} + \ln|3+7x|\right) + C$$

8) Find $\int (3x-5)\cdot e^{x}\cdot dx$ Hint: Use Integration by Parts;

Let $u = 3x-5$ and $dv = e^{x}\cdot dx$

a) $v =$ ____________________

b) $\int (3x-5)\cdot e^{x}\cdot dx$ = ______________________________

Solution:

(8) $\int (3x-5)\cdot e^{x}\cdot dx$

$u = 3x - 5$

$du/dx = 3$

$du = 3\cdot dx$

$\frac{1}{3} du = dx$

$dv = e^{x}\cdot dx$

$\int 1\cdot dv = \int e^{x}\cdot dx$

a) $V = e^{x}$

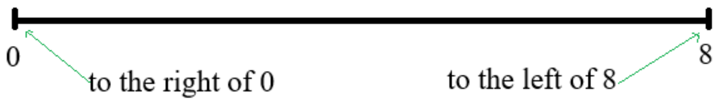
$\int (3x-5)\cdot e^{x}\cdot dx$

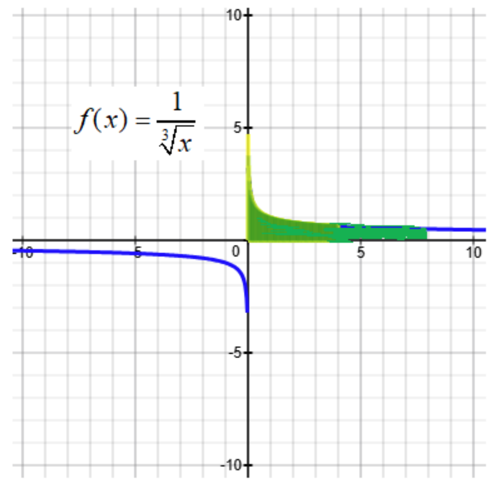
$= \int u\cdot dv = u\cdot v - \int v\cdot du$

$= (3x-5)\cdot e^{x} - \int e^{x}\cdot \cancel{\frac{1}{3}du}$
$3\cdot dx$

(b) $= (3x-5)e^{x} - 3e^{x} + C$

9) Evaluate the improper integral $\int_0^8 \frac{1}{\sqrt[3]{x}}\,dx$. Note: The integrand $\frac{1}{\sqrt[3]{x}}$ is undefined when $x = 0$.





a) Rewrite the improper integral $\int_0^8 \frac{1}{\sqrt[3]{x}}\,dx$ as a limit: $\int_0^8 \frac{1}{\sqrt[3]{x}}\,dx = \lim_{b \to 0^+} \int_?^? \frac{1}{\sqrt[3]{x}}\,dx$

b) Evaluate $\int \frac{1}{\sqrt[3]{x}}\,dx =$ ____?____

c) $\int_0^8 \frac{1}{\sqrt[3]{x}}\,dx =$ ____?____

Solution:

(9) $\int_0^8 \frac{1}{\sqrt[3]{x}}\,dx$

$= \lim_{b \to 0^+} \int_b^8 \frac{1}{\sqrt[3]{x}}\,dx$

$= \lim_{b \to 0^+} \int_b^8 x^{-1/3} \cdot dx$

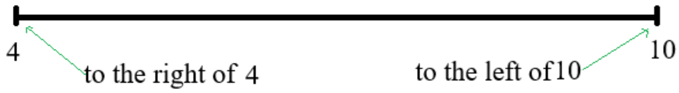
$= \lim_{b \to 0^+} \frac{x^{2/3}}{2/3}$

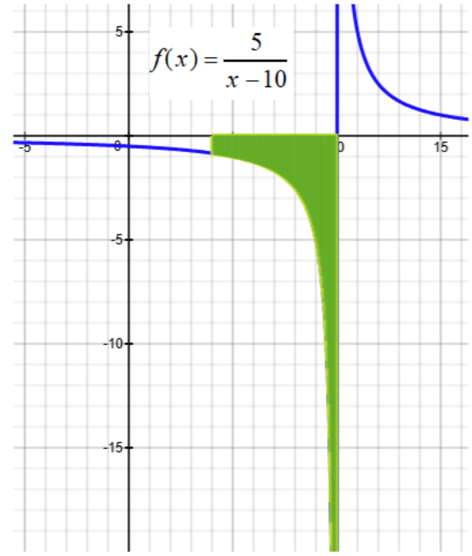
$= \lim_{b \to 0^+} \frac{3}{2} x^{2/3} \Big|_b^8$

$= \lim_{b \to 0^+} \left[\frac{3}{2}(8)^{2/3} - \frac{3}{2}(b)^{2/3} \right]$

$= \frac{3}{2}(4) - \frac{3}{2}(0) = 6$

10) Evaluate the improper integral $\int_4^{10} \frac{5}{x-10}\,dx$ Note: The integrand $\frac{5}{x-10}$ is undefined when $x = 10$.





a) Rewrite the improper integral $\int_4^{10} \frac{5}{x-10}\,dx$ as a limit: $\int_4^{10} \frac{5}{x-10}\,dx = \lim_{b \to 10^-} \int_?^? \frac{5}{x-10}\,dx$

b) Evaluate $\int \frac{5}{x-10}\,dx =$ ___?___

c) $\int_4^{10} \frac{5}{x-10}\,dx =$ ___?___

Solution:

$$\int_4^{10} \frac{5}{x-10}\,dx = \lim_{b \to 10^-} \int_4^b \frac{5}{x-10}\,dx = \lim_{b \to 10^-} \left[5 \cdot \int_4^b \frac{1}{x-10}\,dx \right] = \lim_{b \to 10^-} \left[5 \cdot \ln|x-10| \right]_4^b$$

$$= \lim_{b \to 10^-} \left[5 \cdot \ln|b-10| - 5 \cdot \ln|4-10| \right]$$

$$= \lim_{b \to 10^-} \left[5 \cdot \ln|b-10| - 5 \cdot \ln|-6| \right]$$

Note: As $b \to 10^-$, $|b-10| \to 0$ and $\ln|b-10| \to -\infty$

This means that $\lim_{b \to 10^-} \left[5 \cdot \ln|b-10| - 5 \cdot \ln|-6| \right] = -\infty - 5 \cdot \ln 6 = -\infty$

Therefore, $\int_4^{10} \frac{5}{x-10}\,dx = -\infty$

11) Evaluate the improper integral $\int_5^{\infty} \frac{x}{\sqrt{x^2-4}}\,dx.$

a) Rewrite the improper integral $\int_5^{\infty} \frac{x}{\sqrt{x^2-4}}\,dx$ as a limit: $\int_5^{\infty} \frac{x}{\sqrt{x^2-4}}\,dx = \lim_{b\to\infty} \int_{?}^{?} \frac{x}{\sqrt{x^2-4}}\,dx$

b) Evaluate $\int \frac{x}{\sqrt{x^2-4}}\,dx =$ ____?____

c) $\int_5^{\infty} \frac{x}{\sqrt{x^2-4}}\,dx =$ ____?____

Solution:

$$\int_5^{\infty} \frac{x}{\sqrt{x^2-4}}\,dx = \lim_{b\to\infty} \int_5^{b} \frac{x}{\sqrt{x^2-4}}\,dx$$

Evaluating $\int \frac{x}{\sqrt{x^2-4}}\,dx$

Let $u = x^2 - 4 \quad \Rightarrow \quad \frac{du}{dx} = 2x \quad \Rightarrow \quad du = 2x\,dx \quad \Rightarrow \quad \frac{1}{2}du = x\,dx$

So $\int \frac{x}{\sqrt{x^2-4}}\,dx = \int \frac{1}{\sqrt{u}} \cdot \frac{1}{2}du = \frac{1}{2}\int u^{-1/2}du = \frac{1}{2} \cdot \frac{u^{1/2}}{1/2} = u^{1/2} = \left(x^2-4\right)^{1/2}$

Hence $\lim_{b\to\infty} \int_5^{b} \frac{x}{\sqrt{x^2-4}}\,dx = \lim_{b\to\infty}\left[\left(x^2-4\right)^{1/2}\right]_5^b = \lim_{b\to\infty}\left[\left(b^2-4\right)^{1/2} - \left(5^2-4\right)^{1/2}\right]$

Note: As $b \to \infty$, $\left(b^2-4\right)^{1/2} \to \infty$

This means that $\lim_{b\to\infty}\left[\left(b^2-4\right)^{1/2} - \left(5^2-4\right)^{1/2}\right] = \infty - \left(5^2-4\right)^{1/2} = \infty$

Therefore, $\int_5^{\infty} \frac{x}{\sqrt{x^2-4}}\,dx = \infty$

12) Find $\lim_{n\to\infty} \frac{5n^2}{4n^2+2}$

Solution:

(12) $\lim_{n\to\infty} \frac{5n^2}{4n^2+2}$

$= \lim_{n\to\infty} \frac{5n^2/n^2}{4n^2/n^2 + 2/n^2}$

$= \lim_{n\to\infty} \frac{5}{4+2/n^2} = \frac{5}{4}$

13) Find $\lim\limits_{n\to\infty} \dfrac{n}{\sqrt{n^2+1}}$

Solution:

$$(13)\quad \lim_{n\to\infty} \frac{n}{\sqrt{n^2+1}}$$

$$= \lim_{n\to\infty} \frac{n/\sqrt{n^2}}{\sqrt{n^2+1}\,/\sqrt{n^2}}$$

$$= \lim_{n\to\infty} \frac{n/n}{\sqrt{n^2/n^2 + 1/n^2}}$$

$$= \lim_{n\to\infty} \frac{1}{\sqrt{1 + 1/n^2}}$$

$$= \frac{1}{\sqrt{1+0}} = 1$$

14) Use the nth Term Test to explain why the series $\sum_{n=1}^{\infty} \frac{n}{4n+3}$ converges or diverges.

Solution:

Let $a_n = \frac{n}{4n+3}$

a) $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{4n+3} = \frac{1}{4}$ Using L'Hopital's Rule

b) Since $\lim_{n\to\infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \frac{n}{4n+3}$ diverges by the nth Term Test for Divergence.

15) Use the nth Term Test to explain why the series $\sum_{n=1}^{\infty} \frac{3n}{\sqrt{n^2+1}}$ converges or diverges.

Solution:

Let $a_n = \frac{3n}{\sqrt{n^2+1}}$

a) $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3n}{\sqrt{n^2+1}} = \lim_{n\to\infty} \frac{3n/\sqrt{n^2}}{\sqrt{n^2+1}/\sqrt{n^2}} = \lim_{n\to\infty} \frac{3n/n}{\sqrt{n^2/n^2+1/n^2}} = \lim_{n\to\infty} \frac{3}{\sqrt{1+1/n^2}} = \frac{3}{\sqrt{1+0}} = 3$

b) Since $\lim_{n\to\infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \frac{3n}{\sqrt{n^2+1}}$ diverges by the nth Term Test for Divergence.

16) Use the Geometric Series Test to explain why the series $\sum_{n=0}^{\infty} \left(\frac{6}{5}\right)^n$ converges or diverges.

Solution:

$$\sum_{n=0}^{\infty} \left(\frac{6}{5}\right)^n = \left(\frac{6}{5}\right)^0 + \left(\frac{6}{5}\right)^1 + \left(\frac{6}{5}\right)^2 + \left(\frac{6}{5}\right)^2 \ldots.$$

$$\sum_{n=0}^{\infty} \left(\frac{6}{5}\right)^n = 1 + \left(\frac{6}{5}\right)^1 + \left(\frac{6}{5}\right)^2 + \left(\frac{6}{5}\right)^2 \ldots.$$

$a =$ first term $= 1$ $\qquad$ $r =$ common ratio $= 6/5$

Since $r > 1$, $\sum_{n=0}^{\infty} \left(\frac{6}{5}\right)^n$ diverges by Geometric Series Test.

17) Use the integral test to explain why the series $\sum_{n=1}^{\infty} \frac{5}{3n+2}$ converges or diverges.

Solution:

$$\int_1^{\infty} \frac{5}{3x+2}dx = 5 \cdot \int_1^{\infty} \frac{1}{3x+2}dx = 5 \cdot \left[\frac{1}{3}\ln|3x+2|\right]_1^{\infty} = \frac{5}{3}\left[\ln|3x+2|\right]_1^{\infty}$$

Note: As $x \to \infty$, $\ln|3x+2| \to \infty$.

So $\frac{5}{3}\left[\ln|3x+2|\right]_1^{\infty} = \frac{5}{3}\left[\lim_{x\to\infty} \ln|3x+2| - \ln|3(1)+2|\right] = \frac{5}{3}\left[\infty - \ln|3(1)+2|\right] = \infty$

Therefore, the series $\sum_{n=1}^{\infty} \frac{5}{3n+2}$ diverges by the Integral Test.

18) Use the p-series Test to explain why the series $\sum_{n=1}^{\infty} \frac{3}{n^{1/3}}$ converges or diverges.

Solution:

Since $p = 1/3 < 1$, the series $\sum_{n=1}^{\infty} \frac{3}{n^{1/3}}$ diverges.

19) $\sum_{n=1}^{\infty} \frac{8}{4^n+1}$

Use Limit Comparison Test or Direct Comparison Test to explain why series converges:

A) Compare: $\frac{8}{4^n+1}$ ___?___ $\frac{8}{4^n}$ (Answer should be <, >, or =)

B) $\sum_{n=1}^{\infty} \frac{8}{4^n}$ converges or diverges?

C) $\sum_{n=1}^{\infty} \frac{8}{4^n+1}$ converges or diverges?

Solution:

Let $a_n = \frac{8}{4^n+1}$ and $b_n = \frac{8}{4^n}$

A) Note: $4^n+1 > 4^n$ and $\frac{8}{4^n+1} < \frac{8}{4^n}$ and $\sum_{n=1}^{\infty} \frac{8}{4^n+1} < \sum_{n=1}^{\infty} \frac{8}{4^n}$

B) Next we will show that $\sum_{n=1}^{\infty} \frac{8}{4^n} = 8 \cdot \sum_{n=1}^{\infty} \frac{1}{4^n} = 8 \cdot \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n$ converges to $\frac{8}{3}$.

Note that $\sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n$ is a Geometric Series and $\sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n$ converges to $\frac{a}{1-r} = \frac{1/4}{1-1/4} = \frac{1}{3}$

So $8 \cdot \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n = 8 \cdot \frac{1}{3} = \frac{8}{3}$

C) Therefore, we have $0 < \sum_{n=1}^{\infty} \frac{8}{4^n+1} < \sum_{n=1}^{\infty} \frac{8}{4^n} = \frac{8}{3}$ and $\sum_{n=1}^{\infty} \frac{8}{4^n+1}$ converges by Limit Comparison Test.

20) $\sum_{n=1}^{\infty} \frac{2n+1}{n^3-2n+3}$

Use Limit Comparison Test to explain why series converges or diverges.

Let $a_n \frac{2n+1}{n^3-2n+3}$ and $b_n = \frac{2n}{n^3}$

A) Find $\lim_{x\to\infty}\left(\frac{a_n}{b_n}\right) =$ __?________

B) $\sum_{n=1}^{\infty} \frac{2n}{n^3}$ converges or diverges?

C) $\sum_{n=1}^{\infty} \frac{2n+1}{n^3-2n+3}$ converges or diverges?

Solution:

Let $a_n = \frac{2n+1}{n^3-2n+3}$ and $b_n = \frac{2n}{n^3}$

$$\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = \lim_{n\to\infty}\left(\frac{\frac{2n+1}{n^3-2n+3}}{\frac{2n}{n^3}}\right) = \lim_{n\to\infty}\left(\frac{2n+1}{n^3-2n+3}\cdot\frac{n^3}{2n}\right) = \lim_{n\to\infty}\left(\frac{2n^4+n^3}{2n^4-4n^2+6n}\right) = 1$$

The series $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{2n}{n^3} = 2\cdot\sum_{n=1}^{\infty}\frac{1}{n^2}$ converges by p-series Test.

Therefore, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{2n+1}{n^3-2n+3}$ also converges by Limit Comparison Test.

21) $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{n+7}}$

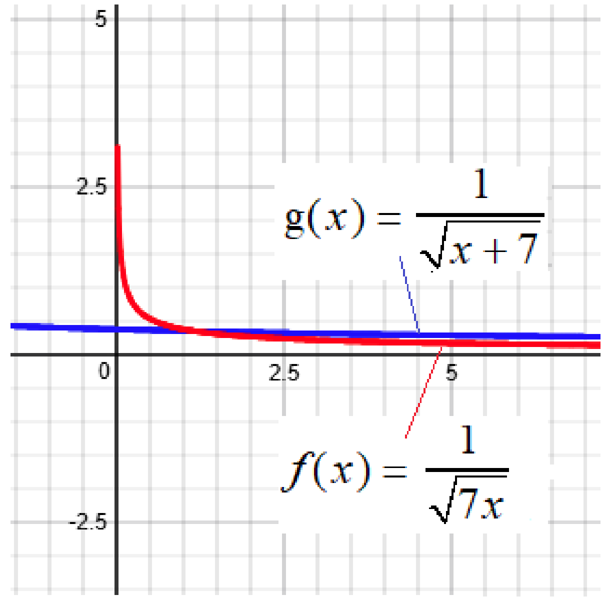
Use the Direct Comparison Test to explain why the series converges or diverges.

Let $a_n = \dfrac{1}{\sqrt{n+7}}$ and $b_n = \dfrac{1}{\sqrt{7n}} = \dfrac{1}{\sqrt{7}}\cdot\dfrac{1}{\sqrt{n}}$

a) Compare $\dfrac{1}{\sqrt{n+7}}$ _?_ $\dfrac{1}{\sqrt{7}}\cdot\dfrac{1}{\sqrt{n}}$

b) $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{7}}\cdot\frac{1}{\sqrt{n}}$ converges or diverges?

c) $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{n+7}}$ converges or diverges?



Solution:

Let $a_n = \dfrac{1}{\sqrt{n+7}}$ and $b_n = \dfrac{1}{\sqrt{7n}}$

a) Note: $\sqrt{n+7} < \sqrt{7n}$ for $n \ge 2$ and $\dfrac{1}{\sqrt{n+7}} > \dfrac{1}{\sqrt{7n}}$ for $n \ge 2$

(We can also see this from the graphs.)

b) The series $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{7n}} = \sum_{n=1}^{\infty}\frac{1}{\sqrt{7}}\cdot\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{7}}\cdot\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$ diverges by p-series Test

because $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty}\frac{1}{n^{1/2}}$ diverges.

c) So we have $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{7n}} = \infty$ and $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{n+7}} > \sum_{n=1}^{\infty}\frac{1}{\sqrt{7n}} = \infty$

Therefore, the series $\displaystyle\sum_{n=1}^{\infty}\frac{1}{\sqrt{n+7}}$ diverges by Directe Comparison Test.

22) $\sum_{n=1}^{\infty} \frac{(-1)^n 4^n}{n!}$

Use Ratio Test to explain why series converges or diverges.

Let $a_n = \frac{(-1)^n 4^n}{n!}$; $a_{n+1} = \frac{(-1)^{n+1} 4^{n+1}}{(n+1)!}$

A) Find $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| =$ _?____

B) $\sum_{n=1}^{\infty} \frac{(-1)^n 4^n}{n!}$ converges or diverges?

Solution:

Let $a_n = \frac{4^n}{n!}$ and $a_{n+1} = \frac{4^{n+1}}{(n+1)!}$

Note: $\frac{n!}{(n+1)!} = \frac{n(n-1)(n-2)\ldots 3\cdot 2\cdot 1}{(n+1)n(n-1)(n-2)\ldots 3\cdot 2\cdot 1} = \frac{1}{n+1}$

A) $$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{\frac{4^{n+1}}{(n+1)!}}{\frac{4^n}{n!}} \right| = \lim_{n\to\infty} \left| \frac{4^{n+1}}{(n+1)!} \cdot \frac{n!}{4^n} \right| = \lim_{n\to\infty} \left| \frac{n!}{(n+1)!} \cdot \frac{4^{n+1}}{4^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{n!}{(n+1)!} \cdot \frac{4^{n+1}}{4^n} \right| = \lim_{n\to\infty} \left| \frac{1}{n+1} \cdot 4 \right| = 0$$

B) Therefore, $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot 4^n}{n!}$ converges by Ratio Test because $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = 0 < 1$.

23) $\sum_{n=1}^{\infty} \frac{3^n}{2^n+1}$

Use Ratio Test to explain why series converges or diverges.

Let $a_n = \frac{3^n}{2^n+1}$; $\quad a_{n+1} = \frac{3^{n+1}}{2^{n+1}+1}$

A) Find $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| =$ _?_____

B) $\sum_{n=1}^{\infty} \frac{3^n}{2^n+1}$ converges or diverges?

Solution:

Let $a_n = \frac{3^n}{2^n+1}$ and $a_{n+1} = \frac{3^{n+1}}{2^{n+1}+1}$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{3^{n+1}}{2^{n+1}+1}}{\frac{3^n}{2^n+1}}\right| = \lim_{n\to\infty}\left|\frac{3^{n+1}}{2^{n+1}+1}\cdot\frac{2^n+1}{3^n}\right| = \lim_{n\to\infty}\left|\frac{3^{n+1}}{3^n}\cdot\frac{2^n+1}{2^{n+1}}\right| = \lim_{n\to\infty}\left|3\cdot\frac{2^n+1}{2^1\cdot 2^n}\right| = 3\cdot\lim_{n\to\infty}\left|\frac{2^n+1}{2\cdot 2^n}\right|$$

Note: $D_x\left(a^x\right) = \frac{1}{\ln a}a^x$

A) By using L'Hopital's Rule, we have: $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 3\cdot\lim_{n\to\infty}\left|\frac{2^n+1}{2\cdot 2^n}\right| = 3\cdot\lim_{n\to\infty}\left|\frac{\frac{1}{\ln 2}\cdot 2^n}{2\cdot\frac{1}{\ln 2}\cdot 2^n}\right| = 3\cdot\lim_{n\to\infty}\left|\frac{1}{2}\right| = \frac{3}{2}$

B) Therefore, $\sum_{n=1}^{\infty}\frac{3^n}{2^n+1}$ diverges by Ratio Test because $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \frac{3}{2} > 1.$

24) $\sum_{n=1}^{\infty}\frac{(-1)^n x^n}{2^n}$

Find interval of convergence for this power series.

Let $u_n = \frac{(-1)^n x^n}{2^n}$ $\quad u_{n+1} = \frac{(-1)^{n+1} x^{n+1}}{2^{n+1}}$

A) Find $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| =$ ___?___

B) Find Interval of Convergence = __________

Solution:

Let $u_n = \frac{(-1)^n x^n}{2^n}$ and $u_{n+1} = \frac{(-1)^{n+1} x^{n+1}}{2^{n+1}}$

A) $$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{(-1)^{n+1} x^{n+1}}{2^{n+1}}}{\frac{(-1)^n x^n}{2^n}}\right| = \lim_{n\to\infty}\left|\frac{x^{n+1}}{x^n}\cdot\frac{2^n}{2^{n+1}}\right| = \lim_{n\to\infty}\left|x\cdot\frac{1}{2}\right| = \left|\frac{1}{2}x\right|$$

Note: The series $\sum_{n=1}^{\infty}\frac{(-1)^n x^n}{2^n}$ converges if $\left|\frac{1}{2}x\right| < 1$

B) So radius of convergence is $\left|\frac{1}{2}x\right| < 1 \iff$ or $-1 < \frac{1}{2}x < 1 \iff -2 < x < 2$

If $x = -2$, $\sum_{n=1}^{\infty}\frac{(-1)^n x^n}{2^n} = \sum_{n=1}^{\infty}\frac{(-1)^n(-2)^n}{2^n} = \sum_{n=1}^{\infty}\frac{2^n}{2^n} = \sum_{n=1}^{\infty}1 = \infty.$

If $x = 2$, $\sum_{n=1}^{\infty}\frac{(-1)^n x^n}{2^n} = \sum_{n=1}^{\infty}\frac{(-1)^n(2)^n}{2^n} = \sum_{n=1}^{\infty}(-1)^n$ diverges.

Therefore, radius of convergence is $(-2, 2)$.

25) $\sum_{n=0}^{\infty} \frac{(5x)^n}{(2n)!}$

Find interval of convergence for this power series.

Let $u_n = \frac{(5x)^n}{(2n)!}$ $\quad u_{n+1} = \frac{(5x)^{n+1}}{(2(n+1))!}$

A) Find $\lim\limits_{n\to\infty} \left|\frac{u_{n+1}}{u_n}\right| =$ ______________________________

B) Find Interval of Convergence = ________________________________

Solution:

Let $u_n = \frac{(5x)^n}{(2n)!}$ $\qquad$ and $\qquad$ $u_{n+1} = \frac{(5x)^{n+1}}{(2(n+1))!}$

A) $$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{(5x)^{n+1}}{(2(n+1))!}}{\frac{(5x)^n}{!}}\right| = \lim_{n\to\infty}\left|\frac{(5x)^{n+1}}{(5x)^n}\cdot\frac{(2n)!}{(2(n+1))!}\right| = \lim_{n\to\infty}\left|5x\cdot\frac{(2n)!}{(2(n+1))!}\right|$$

Note: $$\frac{(2n)!}{(2(n+1))!} = \frac{(2n)!}{(2n+2)!} = \frac{(2n)!}{(2n+2)(2n+1)(2n)!} = \frac{1}{(2n+2)(2n+1)}$$

So $$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = : \lim_{n\to\infty}\left|5x\cdot\frac{(2n)!}{(2(n+1))!}\right| = |5x|\cdot\lim_{n\to\infty}\left|\frac{1}{(2n+2)(2n+1)}\right| = |5x|\cdot 0 = 0.$$

B) Note: The series $\sum_{n=1}^{\infty}\frac{(5x)^n}{(2n)!}$ converges for any x because $\lim\limits_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| < 1$ for any x.

Therefore radius of convergence is $(-\infty, \infty)$.

26) Find the vertex of the following parabola. Then draw graph of parabola.

$y^2 - 12y - 8x + 20 = 0$

vertex=_______?_____________

p =____________?_________________

Solution:

$y^2 - 12y - 8x + 20 = 0$

$y^2 - 12y = 8x - 20$

$y^2 - 12y + 36 = 8x - 20 + 36$

$(y-6)^2 = 8x + 16$

$(y-6)^2 = 8(x+2)$

Matching formula: $(y-k)^2 = 4p(x-h)$

Vertex =(-2, 6)

4p = 8 $\Rightarrow$ p=2

Equations of Parabola:

1) Parabola opens up: $(x-h)^2 = 4p(y-k)$

vertex $=(h,k)$; focus = $(h, k+p)$; directrix: $y = k - p$

2) Parabola opens down: $(x-h)^2 = -4p(y-k)$

vertex $=(h,k)$; focus = $(h, k-p)$; directrix: $y = k + p$

3) Parabola opens right: $(y-k)^2 = 4p(x-h)$

vertex $=(h,k)$; focus = $(h+p, k)$; directrix: $y = h - p$

4) Parabola opens left: $(y-k)^2 = -4p(x-h)$

vertex $=(h,k)$; focus = $(h-p, k)$; directrix: $y = h + p$

27) Find the center and vertices of the following ellipse. Then draw graph of ellipse.

$$\frac{(x-2)^2}{1/5} + y^2 = 1$$

vertices = ________?__________

center = ________?______________

Solution:

$$\frac{(x-2)^2}{1/5} + y^2 = 1 \quad \Leftrightarrow \frac{(x-2)^2}{1/5} + \frac{(y-0)^2}{1} = 1$$

Matching formula: $\dfrac{(x-h)^2}{b^2} + \dfrac{(y-k)^2}{a^2} = 1$

$h = 2; k = 0$ Center$(2, 0)$

$a^2 = 1; \quad a = 1$

$b^2 = 1/5$

$a^2 = b^2 + c^2$

$c^2 = 4/5; \quad c = \sqrt{4/5}$

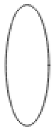
Vertices: $(h, k \pm a) = (2, 0 \pm 1)$

Ellipse Elongated Horizontally 

Equation: $\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1\ ; \quad a > b$

$a^2 = b^2 + c^2$ Center of Ellipse: (h, k)

Vertices: $(h \pm a, k)$ Foci: $(h \pm c, k)$

Ellipse Elongated Vertically 

Equation: $\dfrac{(x-h)^2}{b^2} + \dfrac{(y-k)^2}{a^2} = 1; \quad a > b$

$a^2 = b^2 + c^2$ Center of Ellipse: (h, k)

Vertices: $(h, k \pm a)$ Foci: $(h, k \pm c)$

28) Find the center and vertices of the following hyperbola. Then draw graph of hyperbola.

$$12x^2 - 12y^2 - 12x + 24y - 45 = 0$$

vertices = ______?______

center = ______?______

Hyperbola with Branches Opening Left and Right)(

Equation of Hyperbola: $\dfrac{(x-h)^2}{a^2} - \dfrac{(y-k)^2}{b^2} = 1$

$c^2 = a^2 + b^2$ Center of Hyperbola: (h,k)

Vertices: $(h \pm a, k)$ Foci: $(h \pm c, k)$

Equations of asymptotes: $y = \pm \dfrac{b}{a}(x-h) + k$

Hyperbola with Branches Opening Up and Down

Equation of Hyperbola: $\dfrac{(y-k)^2}{a^2} - \dfrac{(x-h)^2}{b^2} = 1$

$c^2 = a^2 + b^2$ Center of Hyperbola: (h,k)

Vertices: $(h, k \pm a)$ Foci: $(h, k \pm c)$

Equations of asymptotes: $y = \pm \dfrac{a}{b}(x-h) + k$

Solution:

$$12x^2 - 12y^2 - 12x + 24y - 45 = 0$$

$$(12x^2 - 12x) + (-12y^2 + 24y) = 45$$

$$12(x^2 - 1x) - 12(y^2 - 2y) = 45$$

half of 1 = 1/2 half of -2 = -1

$(1/2)^2 = 1/4$ and $(-1)^2 = 1$

$$12(x^2 - x + 1/4) - 12(y^2 - 2y + 1) = 45 + 12(1/4) - 12(1)$$

$$12(x^2 - x + 1/4) - 12(y^2 - 2y + 1) = 36$$

$$\frac{(x-1/2)^2}{3} - \frac{(y-1)^2}{3} = 1$$

Matching formula: $\dfrac{(x-h)^2}{a^2} - \dfrac{(y-k)^2}{b^2} = 1$

h = 1/2 and k = 1

$a^2 = 3$; $b^2 = 3$

$a = \sqrt{3}$; $b = \sqrt{3}$

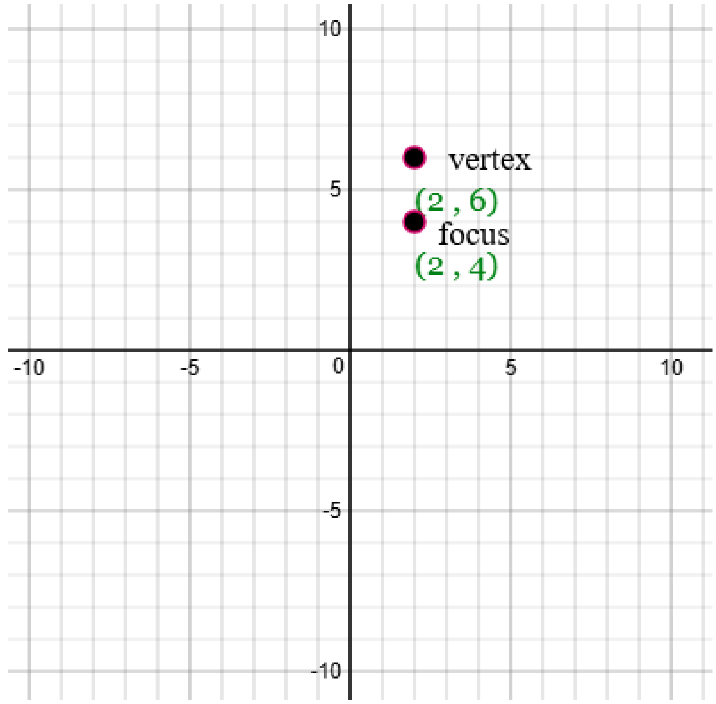
Center = (1/2, 1)

vertices = $(1/2 \pm \sqrt{3},\ 1)$

29) Find equation of parabola.

vertex = (2, 6) focus = (2, 4)

Equation of parabola:___________________



Solution:

Parabola opens down.

p = distance from vertex to focus = 2

$(x - h)^2 = -4p(y-k)$

$(x - 2)^2 = -4(2)(y-6)$

Equations of Parabola:

1) Parabola opens up: $(x-h)^2 = 4p(y-k)$

vertex $= (h, k)$; focus $= (h, k+p)$; directrix: $y = k - p$

2) Parabola opens down: $(x-h)^2 = -4p(y-k)$

vertex $= (h, k)$; focus $= (h, k-p)$; directrix: $y = k + p$

3) Parabola opens right: $(y-k)^2 = 4p(x-h)$

vertex $= (h, k)$; focus $= (h+p, k)$; directrix: $x = h - p$

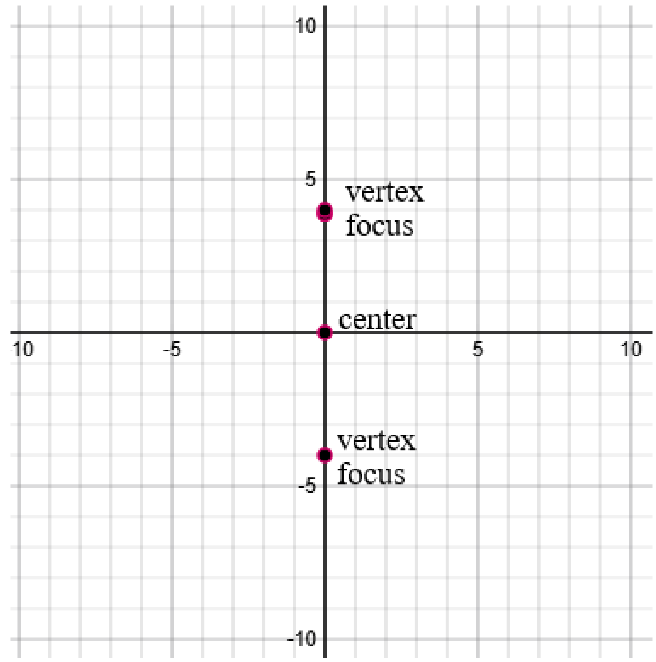
4) Parabola opens left: $(y-k)^2 = -4p(x-h)$

vertex $= (h, k)$; focus $= (h-p, k)$; directrix: $x = h + p$

30) Find equation of ellipse.

center = (0, 0) foci = $(0, \pm\sqrt{15})$ vertices = $(0, \pm 4)$

Equation of ellipse:______________________



Solution:

a = distance from center to vertex =4

$c =$ distance from center to focus = $\sqrt{15}$

$a^2 = 16;$ $c^2 = 15$

$a^2 = b^2 + c^2$

$b^2 = 1$

Matching formula: $\dfrac{(x-h)^2}{b^2} + \dfrac{(y-k)^2}{a^2} = 1$

$$\frac{(x-0)^2}{1} + \frac{(y-0)^2}{16} = 1$$

Note: Ellipse is of vertical type; thus, a^2 is placed under $(y - k)^2$.

Ellipse Elongated Horizontally

Equation: $\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1$; $a > b$

$a^2 = b^2 + c^2$ Center of Ellipse: (h,k)

Vertices: $(h \pm a, k)$ Foci: $(h \pm c, k)$

Ellipse Elongated Vertically

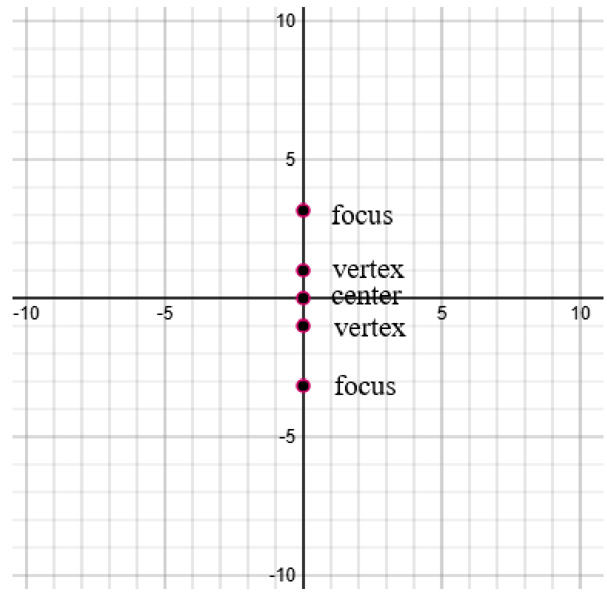
Equation: $\dfrac{(x-h)^2}{b^2} + \dfrac{(y-k)^2}{a^2} = 1;$ $a > b$

$a^2 = b^2 + c^2$ Center of Ellipse: (h,k)

Vertices: $(h, k \pm a)$ Foci: $(h, k \pm c)$

31) Find equation of hyperbola.

center = (0, 0) foci = $(0, \pm\sqrt{10})$ vertices = $(0, \pm 1)$



Equation of ellipse:_________________________________

Solution:

a = distance from center to vertex =1

$c =$ distance from center to focus = $\sqrt{10}$

$a^2 = 1;$ $c^2 = 10$

$c^2 = a^2 + b^2$

$b^2 = 9$

Matching formula: $\dfrac{(y-k)^2}{a^2} - \dfrac{(x-h)^2}{b^2} = 1$

$$\frac{(y-0)^2}{1} - \frac{(x-0)^2}{9} = 1$$

Note: Branches of hyperbola opens up and down.

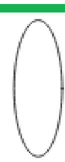
Ellipse Elongated Horizontally

Equation: $\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1$; $a > b$

$a^2 = b^2 + c^2$ Center of Ellipse: (h, k)

Vertices: $(h \pm a, k)$ Foci: $(h \pm c, k)$

Ellipse Elongated Vertically



Equation: $\dfrac{(x-h)^2}{b^2} + \dfrac{(y-k)^2}{a^2} = 1;$ $a > b$

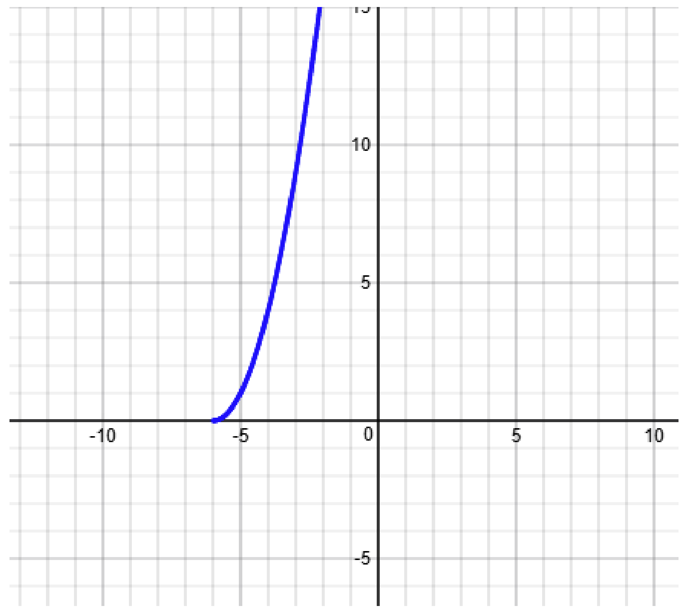
$a^2 = b^2 + c^2$ Center of Ellipse: (h, k)

Vertices: $(h, k \pm a)$ Foci: $(h, k \pm c)$

32) For the following parametric equations, find a corresponding rectangular equation. Then draw graph.

$x = t - 6, \quad y = t^2$

Rectangular Equation:____________________



Solution:

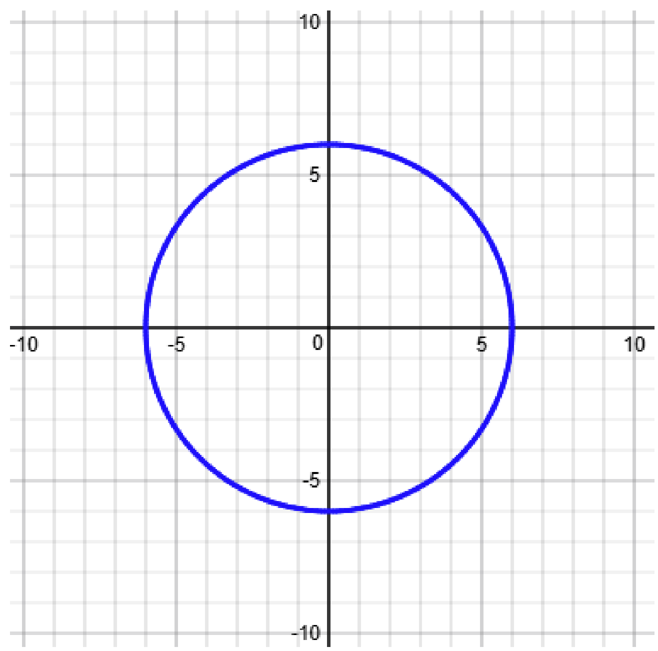
$x = t - 6 \Rightarrow t = x + 6$

$y = t^2 \Rightarrow y = (x+6)^2$ Rectangular Equation

33) For the following parametric equations, find a corresponding rectangular equation. Then draw graph.

$x = 6\cos t \quad y = 6\sin t$

Rectangular Equation:____________________



Solution:

$x = 6\cos t \quad \Rightarrow \cos t = \dfrac{x}{6}$

$y = 6\sin t \quad \Rightarrow \sin t = \dfrac{y}{6}$

From Trigonometry: $\cos^2 t + \sin^2 t = 1$

$\left(\dfrac{x}{6}\right)^2 + \left(\dfrac{y}{6}\right)^2 = 1$ Rectangular Equation

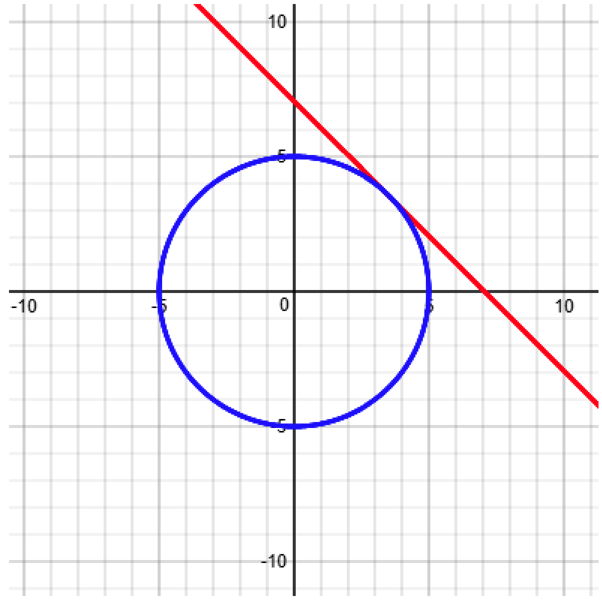
34) For the following parametric equations, find $y' = dy/dx$.

Parametric equations: $x = 5\cos t$ and $y = 5\sin t$

a) $\dfrac{dy}{dx} = \underline{\quad ? \quad}$

b) Equation of tangent line at the point $\left(\dfrac{5\sqrt{2}}{2}, \dfrac{5\sqrt{2}}{2}\right)$ is $\underline{\ ?\ }$

c) Graph $x = 5\cos t$ and $y = 5\sin t$ and tangent line.



Solution:

$$\frac{dy}{dx} = \frac{5\cos t}{-5\sin t} = -\frac{\cos t}{\sin t} = -\cot t$$

At the point $\left(\dfrac{5\sqrt{2}}{2}, \dfrac{5\sqrt{2}}{2}\right)$, $x = \dfrac{5\sqrt{2}}{2}$ and $y = \dfrac{5\sqrt{2}}{2}$

$$y = 5\sin t \quad \Rightarrow \quad \frac{5\sqrt{2}}{2} = 5\sin t \quad \Rightarrow \quad \sin t = \frac{\sqrt{2}}{2} \quad \Rightarrow \quad t = \frac{\pi}{4}$$

Slope of tangent line at the point $\left(\dfrac{5\sqrt{2}}{2}, \dfrac{5\sqrt{2}}{2}\right)$ is $\dfrac{dy}{dx} = -\cot t = -\cot\dfrac{\pi}{4} = -1$

Equation of tangent line is $y - y_1 = m(x - x_1)$

$$y - \frac{5\sqrt{2}}{2} = -1\left(x - \frac{5\sqrt{2}}{2}\right)$$

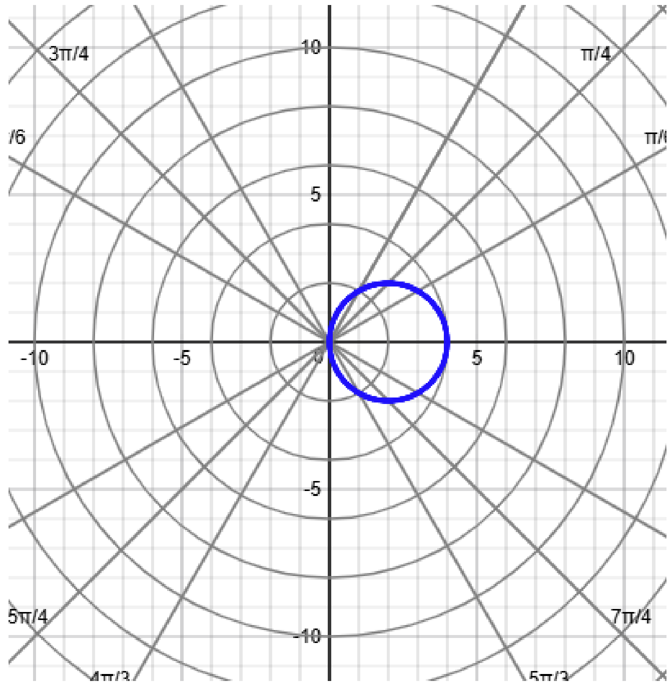
$$y = -1\left(x - \frac{5\sqrt{2}}{2}\right) + \frac{5\sqrt{2}}{2}$$

35) For the following polar equation, find a corresponding rectangular equation and draw graph.

Polar equation: $r = 4\cos\theta$

a) Rectanular equation:___?___

b) Graph $r = 4\cos\theta$



Solution:

$r = 4\cos\theta$ polar equation

$r \cdot r = r \cdot 4\cos\theta$

$r^2 = 4r\cos\theta$

$x^2 + y^2 = 4x$ rectangular equation

36) For the following polar equation, find a corresponding rectangular equation and draw graph

Polar equation: $r = \dfrac{1}{2 - \cos\theta}$

a) Rectangular equation:___?___

b) Graph $r = 4\cos\theta$

Solution:

$$r = \frac{1}{2 - \cos\theta} \qquad \text{polar equation}$$

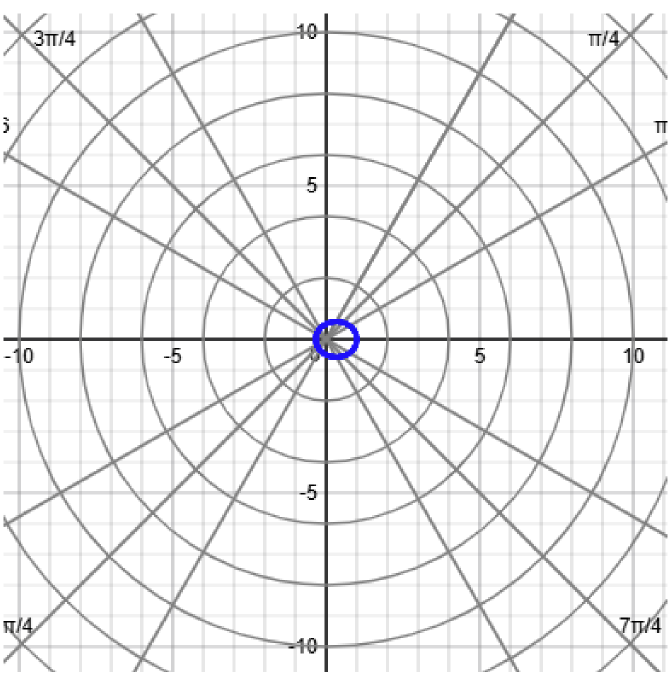
$$r(2 - \cos\theta) = \frac{1}{(2 - \cos\theta)} \cdot (2 - \cos\theta)$$

$$2r - r\cos\theta = 1$$

$$2\sqrt{x^2 + y^2} - x = 1$$

$$\sqrt{x^2 + y^2} = \frac{1 + x}{2}$$

$$(x^2 + y^2)^2 = \left(\frac{1 + x}{2}\right)^2 \qquad \text{rectangular equation}$$

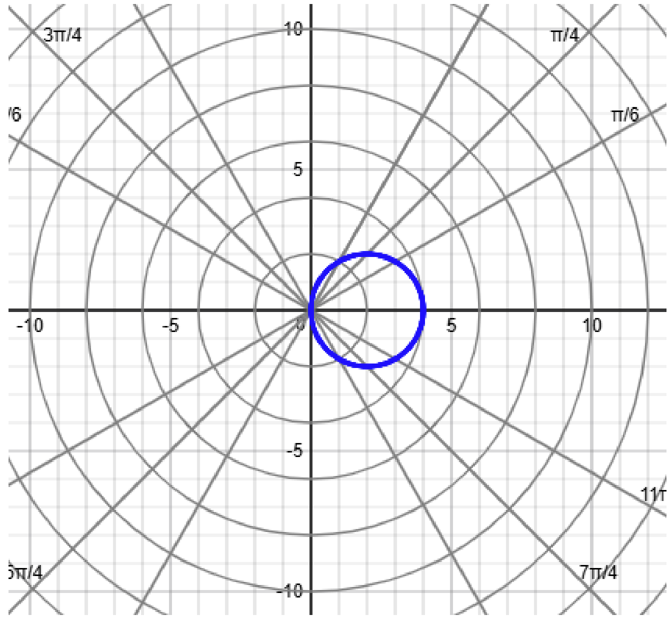


37) For the following rectangular equation, find a corresponding polar equation and draw graph.

Rectangular equation: $x^2 + y^2 - 4x = 0$

a) Polar equation:___?___

b) Graph $x^2 + y^2 - 4x = 0$



Solution:

$x^2 + y^2 - 4x = 0$ rectangular equation

$r^2 - 4r\cos\theta = 0$

$r - 4\cos\theta = 0$

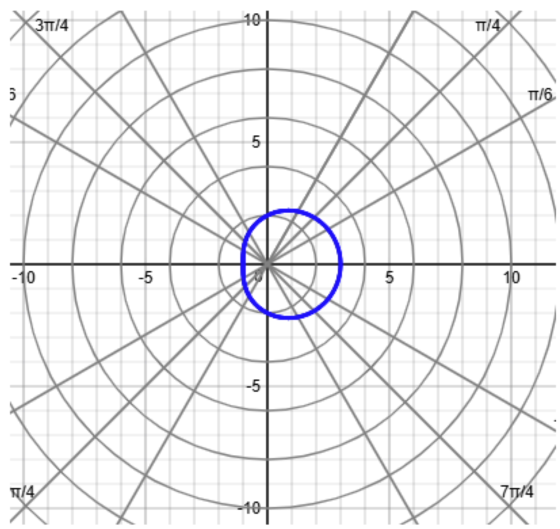
$r = 4\cos\theta$ polar equation

38) Find the area bounded by the following polar equation.

Polar equation: $r = 2 + \cos\theta \quad 0 \le \theta \le 2\pi$

a) Graph $r = 2 + \cos\theta \quad 0 \le \theta \le 2\pi$

b) Area of bounded region = ___?___



Solution:

$$\text{Area} = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \cdot d\theta = \frac{1}{2}\int_{0}^{2\pi} (2+\cos\theta)^2 \cdot d\theta = \frac{9\pi}{2} = 14.13716694$$

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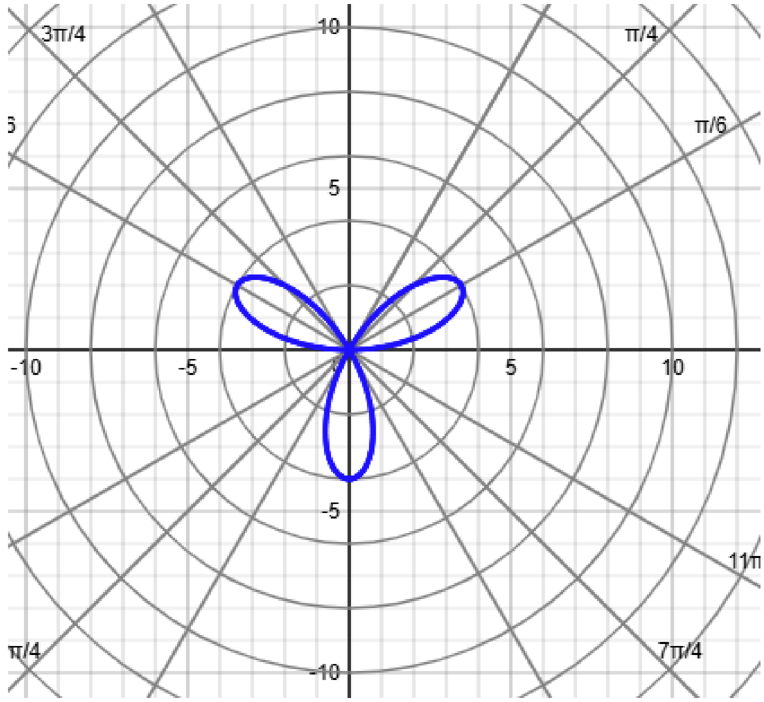
39) Find the area of one petal.

Polar equation: $r = 4\sin 3\theta \quad 0 \le \theta \le \pi$

a) Graph $\quad r = 4\sin 3\theta \qquad 0 \le \theta \le \pi$

b) Area of one petal = ___?___

Hint: Set $r = 0$ and find θ.



Solution:

$r = 4\sin 3\theta$

$\sin 3\theta = r/4$

$3\theta = \sin^{-1}(r/4)$

The petal in Quadrant I starts and ends at the pole.

At the pole, $r = 0$.

$3\theta = \sin^{-1}(r/4) = \sin^{-1}(0/4) = \sin^{-1}(0) = 0, \ \ \pi, \ 2\pi, \ldots$

$\theta = 0, \ \ \pi/3, \ 2\pi/3, \ldots$

Hence the petal in Quadrant I starts at $\theta = 0$ and ends at $\theta = \pi/3$

Therefore,

$$\text{Area of one petal} = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \cdot d\theta = \frac{1}{2}\int_{0}^{\pi/3} (4\sin 3\theta)^2 \cdot d\theta = \frac{4\pi}{3} = 4.1887902047$$

40) Polar equation: $r = \dfrac{6}{3+2\cos\theta}$ $\quad 0 \le \theta \le 2\pi$

a) Multiplying by 1/3: $r = \dfrac{(1/3)\cdot 6}{(1/3)\cdot 3+(1/3)\cdot 2\cos\theta} =$ __?__

b) $e =$ eccentricity = __?__

c) Corresponding rectangular equation: __?__

Hint: Multiply both sides of $r = \dfrac{6}{3+2\cos\theta}$ by $3+2\cos\theta$.

Solution:

Polar equation: $r = \dfrac{6}{3+2\cos\theta}$

a) Multiplying by 1/3: $r = \dfrac{(1/3)\cdot 6}{(1/3)\cdot 3+(1/3)\cdot 2\cos\theta} = \dfrac{2}{1+\frac{2}{3}\cos\theta}$

b) $e =$ eccentricity $= \dfrac{2}{3}$

c) Corresponding rectangular equation:

$r = \dfrac{6}{3+2\cos\theta}$ polar equation

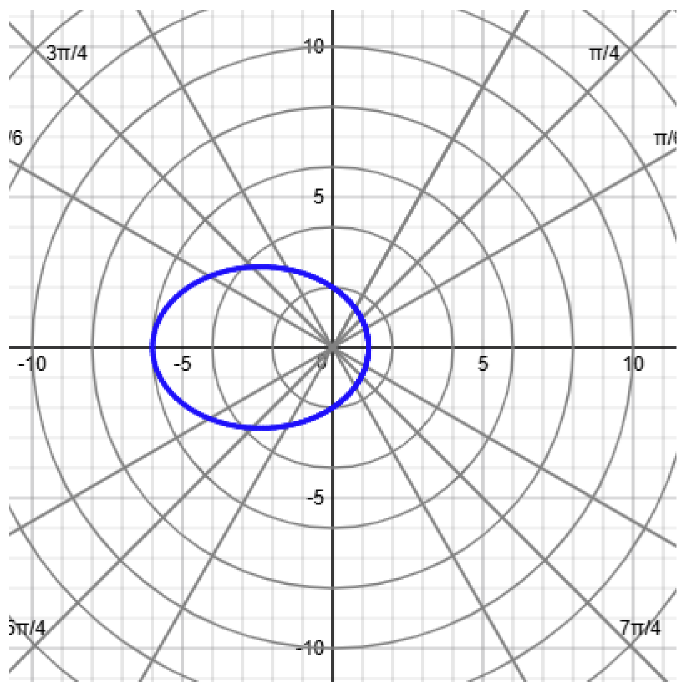
$r(3+2\cos\theta) = \dfrac{6}{3+2\cos\theta}\cdot(3+2\cos\theta)$

$3r + 2r\cos\theta = 6$

$3\sqrt{x^2+y^2} + 2x = 6$

$\sqrt{x^2+y^2} = \dfrac{6-2x}{3}$

$x^2+y^2 = \left(\dfrac{6-2x}{3}\right)^2$ rectangular equation



For #41 and 42

Four types of equations based on location and type of directrix:

1) Horizontal directrix above the pole: $r = \dfrac{ed}{1 + e\sin\theta}$

2) Horizontal directrix below the pole: $r = \dfrac{ed}{1 - e\sin\theta}$

3) Vertical directrix to the right of the pole: $r = \dfrac{ed}{1 + e\cos\theta}$

4) Vertical directrix to the left of the pole: $r = \dfrac{ed}{1 - e\cos\theta}$

41) Find a polar equation for the ellipse with its focus at the pole.

Ellipse has: e = eccentricity = 1/3; Directrix: $y = 4$

a) focus =__________

b) Find distance from focus at the pole to directrix $= |d| =$ ___?___

c) Polar equation for ellipse: ___?_____

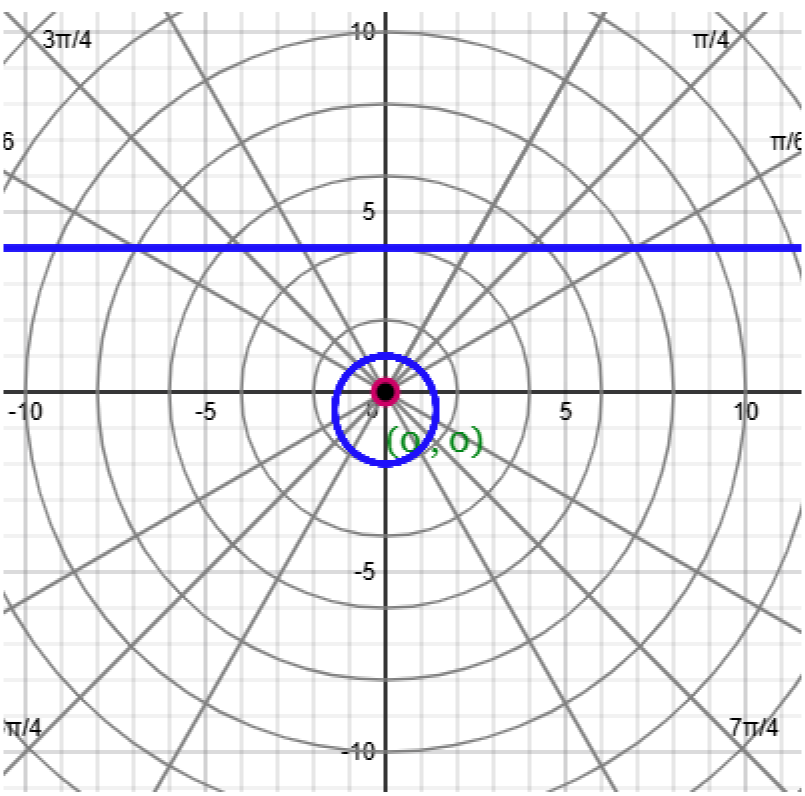
Solution:

Directrix is horizontal and above the pole.

Equation of ellipse is $r = \dfrac{e \cdot d}{1 + e\sin\theta}$.

$d = 4$

$$r = \frac{(1/3)\cdot 4}{1 + \frac{1}{3}\sin\theta} = \frac{4/3}{1 + \frac{1}{3}\sin\theta}$$



42) Find a polar equation for the hyperbola with its focus at the pole.

$e = 5$ and directrix: $x = 3$.

Hint: d = distance from focus (or pole) to the directrix.

Solution:

Directrix is vertical and to the right of the pole.

Equation of hyperbola is $r = \dfrac{e \cdot d}{1 + e\cos\theta}$.

$d = 3$

$$r = \frac{5 \cdot 3}{1 + 5\cos\theta} = \frac{15}{1 + 5\cos\theta}$$